CHAPTER 4

The Network Layer: Data Plane

We learned in the previous chapter that the transport layer provides various forms of process-to-process communication by relying on the network layer's host-to-host communication service. We also learned that the transport layer does so without any knowledge about how the network layer actually implements this service. So perhaps you're now wondering, what's under the hood of the host-to-host communication service, what makes it tick?

In this chapter and the next, we'll learn exactly how the network layer can provide its host-to-host communication service. We'll see that unlike the transport and application layers, *there is a piece of the network layer in each and every host and router in the network.* Because of this, network-layer protocols are among the most challenging (and therefore among the most interesting!) in the protocol stack.

Since the network layer is arguably the most complex layer in the protocol stack, we'll have a lot of ground to cover here. Indeed, there is so much to cover that we cover the network layer in two chapters. We'll see that the network layer can be decomposed into two interacting parts, the **data plane** and the **control plane**. In Chapter 4, we'll first cover the data plane functions of the network layer—the *per-router* functions in the network layer that determine how a datagram (that is, a network-layer packet) arriving on one of a router's input links is forwarded to one of that router's output links. We'll cover both traditional IP forwarding (where forwarding is based on a datagram's destination address) and generalized forwarding (where forwarding and other functions may be performed using values in several different fields in the datagram's header). We'll study the IPv4 and IPv6 protocols and addressing in detail. In Chapter 5, we'll cover the control plane functions of the network layer—the *network-wide* logic that controls how a datagram is routed